

Electric Circuit Analysis-I			
Course Code	EE	Credits	3
Hours/Week (L:T:P:S)	3:0:0:0	CIE Marks	50
Total Hours	45	SEE Marks	50
Exam Hours	3	Course Type	ESC

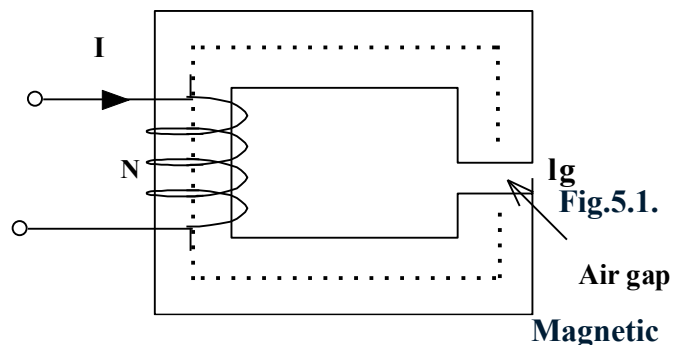
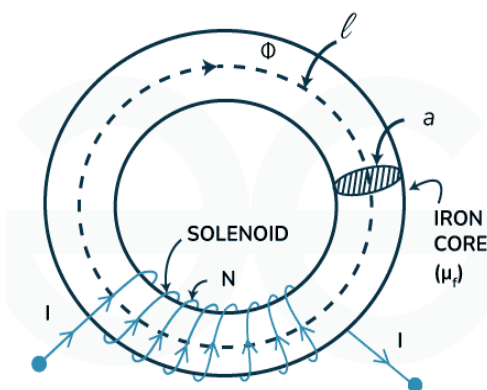
Unit-5	9 Hours
Concept of Magnetic Circuits	
Statically induced emf (Self and Mutually induced emf) (transformer concepts), Dynamically induced emf (DC generator, motor concepts).	
Coupled circuits: Coefficient of coupling, dot convention for coupled coils and analysis of simple coupled circuits.	

5.1. Concept of magnetic circuits

Introduction to magnetic circuits:

A magnetic circuit is a closed path followed by magnetic flux, analogous to an electric circuit's path for electric current. It consists of magnetic materials of high permeability like iron or steel, often with air gaps, and is used to direct and control magnetic fields. These circuits are found in devices like electric motors, transformers, and relays.

Consider the magnetic circuit shown in Fig.5.1



circuits

N = number of turns

a = area of cross section in m^2

l = circumferential length in m

Φ = magnetic flux in Wb

I = current in A

The operation of a magnetic circuit relies on the principle that a current flowing through a coil creates a magnetomotive force (MMF), which drives magnetic flux through the circuit. The opposition to this flux is called reluctance, analogous to resistance in an electrical circuit.

Terms used in magnetic circuit

- **Magnetic Core:**

The primary material within the magnetic circuit, often made of iron or steel that provides a path for the magnetic flux. Examples: Air, Iron, Ferrite, Copper

- **Air Gap:**

A non-magnetic space within the magnetic circuit, often introduced to control the magnetic field or create a specific magnetic effect.

Magnetic Field: A magnetic field is a region around a magnet. Magnetic field lines are used to visualize the field.

Magnetic field strength: It is often denoted by H , is a measure of the magnetizing force that produces a magnetic field in a material. It's a vector quantity, meaning it has both magnitude and direction. The SI unit for magnetic field strength is the ampere per meter (A/m).

- **Magnetomotive Force (MMF):** The driving force that creates magnetic flux, analogous to electromotive force (EMF) in an electrical circuit. It is generated by a current (I) flowing through a coil with a certain number of turns (N), expressed as $MMF = I * N$.

The unit of MMF is ampere-turns (AT).

- **Magnetic Flux:** The magnetic field lines existing in a magnetic field is the magnetic flux Φ . It is analogous to current in an electrical circuit. It's directly proportional to the MMF and inversely proportional to the reluctance. Its unit is weber and symbol is Wb.

- **Magnetic flux density:** It is defined as the amount of magnetic flux passing through a unit area perpendicular to the magnetic field lines (B). It is also known as magnetic induction, is a measure of the strength and direction of a magnetic field. It is defined as the magnetic flux per unit area. $\left(B = \frac{\Phi}{a}\right)$. Its unit is Wb/m².

- **Reluctance:** The opposition to the establishment of magnetic flux in a magnetic circuit is called Reluctance of the circuit $\mathcal{R}$. It is analogous to resistance in an electrical circuit. Reluctance depends on the material's permeability (μ), the length (l) of the magnetic path, and the cross-sectional area (a) of the path. Its unit is Ampere-turns per Weber (AT/Wb).

Reluctance of the magnetic circuit can be calculated using the equation $\mathcal{R} = \frac{l}{\mu a}$

- **Permeability:** Permeability (μ) refers to the ability of a material to allow magnetic field to pass through it. It is similar to conductivity in electrical circuits. It is given by $\mu = \mu_0 \mu_r$

μ_0 = absolute permeability ($4\pi \times 10^{-7}$ H/m (Henry per meter)

μ_r = relative permeability (free space (vacuum)=1, Copper=0.999, wood=1, Iron=5500)

It is also defined as the ratio of Magnetic Flux Density to Magnetic Field Strength.

$$\mu = \frac{B}{H}$$

Analogy between Magnetic and Electric Circuits:

Sl.No	Magnetic Circuit	Electric Circuit
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1	MMF in AT	EMF in Volts
2	Flux exists (Wb)	Current flows (A)
3	Flux Density ($B = \frac{\phi}{a}$)	Current Density ($J = \frac{I}{a}$)
4	Reluctance ($\mathcal{R} = \frac{\ell}{\mu a}$)	Resistance ($R = \zeta \frac{\ell}{a}$)
5	Permeance ($P = \frac{1}{\mathcal{R}}$)	Conductance ($G = \frac{1}{R}$)

Applications:

- Magnetic circuits are fundamental to the operation of various electrical and electromechanical devices.

Examples include:

- **Electric motors:** Magnetic circuits create the magnetic fields that drive the rotation of the motor's rotor. Ex: DC Motor, Induction Motor
- **Transformers:** Magnetic circuits transfer electrical energy between circuits through magnetic coupling.
- **Relays:** Magnetic circuits use magnetic force to control the switching of electrical circuits.
- **Generators:** Magnetic circuits are used to induce voltage by moving conductors within a magnetic field. Ex: DC generator, Synchronous Generator
- **Galvanometers:** Magnetic circuits are used to measure electric current.

In essence, magnetic circuits are the pathways through which magnetic fields travel, and understanding them is crucial for designing and analysing a wide range of electromagnetic devices.

5.2 Analysis of Series Magnetic Circuits:

In a series magnetic circuit, the same flux ϕ flows through each part of the circuit.

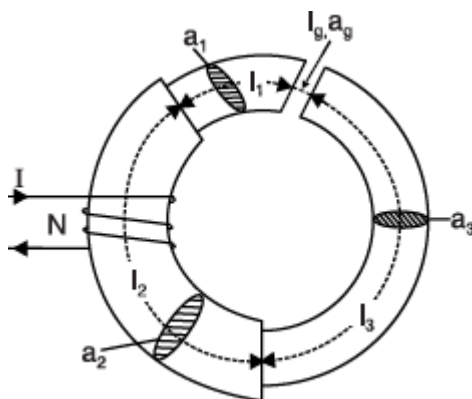


Fig.5.2. Series magnetic circuit

Consider a ***composite series magnetic circuit** consisting of three different magnetic materials of different relative permeability along with an air gap as shown in Fig.5.2. Each part of this series magnetic circuit will offer reluctance to the magnetic flux Φ . The reluctance offered by each part will depend upon dimensions and μ_r of that part. Since the different parts of the circuit are in series, the total reluctance is equal to the sum of reluctances of individual parts, *i.e.*

Step-1: To determine reluctance and MMF of each material

Reluctance of material-1: $\mathcal{R}_1 = \frac{l_1}{a_1 \mu_0 \mu_{r1}}$

Magnetomotive Force (MMF1) of material -1: $\text{MMF}_1 = \phi \times \mathcal{R}_1 = \phi \left[\frac{l_1}{a_1 \mu_0 \mu_{r1}} \right]$ (5.1)

Reluctance of material-2: $\mathcal{R}_2 = \frac{l_2}{a_2 \mu_0 \mu_{r2}}$

Magnetomotive Force (MMF2) of material-2: $\text{MMF}_2 = \phi \times \mathcal{R}_2 = \phi \left[\frac{l_2}{a_2 \mu_0 \mu_{r2}} \right]$ (5.2)

Reluctance of material-3: $\mathcal{R}_3 = \frac{l_3}{a_3 \mu_0 \mu_{r3}}$

Magnetomotive Force (MMF3) of material-3: $\text{MMF}_3 = \phi \times \mathcal{R}_3 = \phi \left[\frac{l_3}{a_3 \mu_0 \mu_{r3}} \right]$ (5.3)

Reluctance of air gap: $\mathcal{R}_g = \frac{l_g}{a_g \mu_0 \mu_{rg}} = \frac{l_g}{a_g \mu_0 (1)} = \frac{l_g}{a_g \mu_0}$ ($\mu_{rg} = 1$)

Magnetomotive Force (MMF-g) of air gap: $\text{MMF}_g = \phi \times \mathcal{R}_g = \phi \left[\frac{l_g}{a_g \mu_0} \right]$ (5.4)

Step-2: To determine total MMF

Total MMF = $\text{MMF}_1 + \text{MMF}_2 + \text{MMF}_3 + \text{MMF}_g$ (5.5)

Substituting equation (5.1) to (5.4) in (5.5)

Total MMF = $\phi \left[\frac{l_1}{a_1 \mu_0 \mu_{r1}} \right] + \phi \left[\frac{l_2}{a_2 \mu_0 \mu_{r2}} \right] + \phi \left[\frac{l_3}{a_3 \mu_0 \mu_{r3}} \right] + \phi \left[\frac{l_g}{a_g \mu_0} \right]$ (5.6)

Exercise 1: An iron ring of cross-sectional area 6 cm^2 is wound with a wire of 100 turns and has a saw cut of 2 mm as in Fig.5.3 Calculate the magnetising current required to produce a flux of 0.1 mWb if mean length of magnetic path is 30 cm and relative permeability of iron is 470.

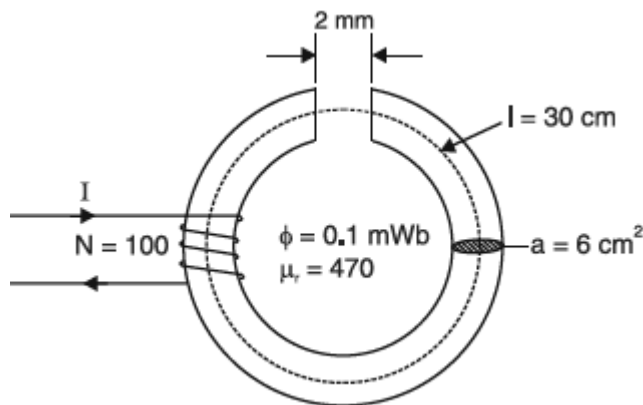


Fig.5.3: Exercise 1

Solution:

Step-1: Note down given data

Number of turns = $N = 100$

Length of iron = $l_i = 30 \times 10^{-2} \text{ m}$

Flux = $\Phi = 0.1 \text{ mWb}$

Area of cross section of iron = $a_i = 6 \times 10^{-4} \text{ m}^2$

Relative Permeability for the air gap = $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$

Relative Permeability for iron = $\mu_r = 470$

Length of air gap= $\ell_g=2 \times 10^{-3}$ m

Area of cross section of air gap= $a_i=6 \times 10^{-4}$ m²

Relative Permeability for air gap= $\mu_0=1$

Step-2: Calculate MMF require for the iron and airgap

$$\text{MMF required for iron material} = \oint \left[\frac{\ell_i}{a_i \mu_0 \mu_{r_i}} \right] = 0.1(10^{-3}) \left[\frac{30 \times 10^{-2}}{6 \times 10^{-4} (4\pi \times 10^{-7}) \times 470} \right] = 84.83 \text{ AT}$$

$$\text{MMF require for air gap} = \oint \left[\frac{\ell_g}{a_g \mu_0 \mu_{r_g}} \right] = 0.1(10^{-3}) \left[\frac{2 \times 10^{-3}}{6 \times 10^{-4} (4\pi \times 10^{-7}) \times 1} \right] = 3.51 \text{ AT}$$

Step-3: Calculate total MMF

Total MMF=MMF required for iron material+ MMF require for air gap= $84.83+3.51=350.63$ AT

Step-4: Determine magnetizing current required

$$NI=350.63$$

$$100I=350.63$$

$$I=350.63/100$$

$$I=3.50 \text{ A}$$

5.3 Analysis of Parallel Magnetic Circuits:

A magnetic circuit which has more than one path for flux is called a **parallel magnetic circuit**.

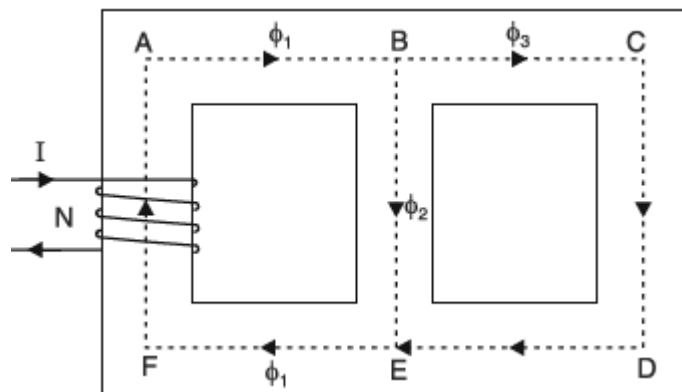


Fig.5.4.Parallel magnetic circuit

It can just be compared to a parallel electric circuit which has more than one path for electric current. The concept of parallel magnetic circuit is illustrated in Fig.5.4. Here a coil of N turns is wound on limb AF carries a current of I amperes. The flux Φ_1 set up by the coil divides at B into two paths, namely:

(i) flux Φ_2 passes along the path BE

(ii) flux Φ_3 follows the path BCDE

$$\text{Clearly, } \Phi_1 = \Phi_2 + \Phi_3$$

The magnetic paths BE and BCDE are in parallel and form a parallel magnetic circuit. The AT required for this parallel circuit is equal to AT required for any one of the paths.

Let

$\mathcal{R}_1$ = reluctance of path EFAB

$\mathcal{R}_2$ = reluctance of path BE

$\mathcal{R}_3$ = reluctance of path BCDE

Total m.m.f. required = m.m.f. for path EFAB + m.m.f. for path BE or path BCDE

$$NI = \Phi_1 \mathcal{R}_1 + \Phi_2 \mathcal{R}_2$$

or

$$NI = \Phi_1 \mathcal{R}_1 + \Phi_3 \mathcal{R}_3$$

The reluctances $\mathcal{R}_1$, $\mathcal{R}_2$ and $\mathcal{R}_3$ must be determined from a calculation of $l/a\mu_0\mu_r$ for those paths of the magnetic circuit in which Φ_1 , Φ_2 and Φ_3 exist respectively.

Exercise 2:

A cast steel magnetic structure made of a bar of section $2 \text{ cm} \times 2 \text{ cm}$ is shown in Fig. 5.5. Determine the current that the 500 turn magnetising coil on the left limb should carry so that a flux of 2 mWb is produced in the right limb. Take $\mu_r = 600$ and neglect leakage.

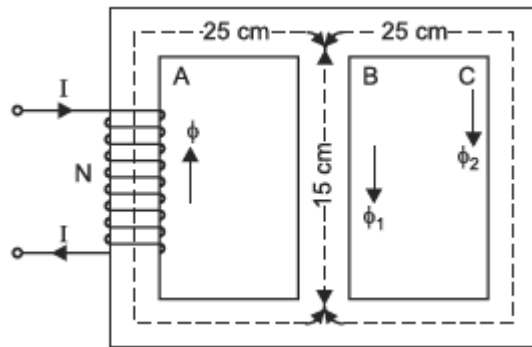


Fig.5.5 Exercise 2

Solution:

Step-1: Note the data given

Area of cross-section = $a = 2 \text{ cm} \times 2 \text{ cm} = 4 \text{ cm}^2$

Number of turns = $N = 500$

Flux in right limb = $\Phi = 2 \text{ mWb}$

$\mu_r = 600$

Step-2: To calculate total flux required

MMF required for the middle section $15 \text{ cm} = \text{MMF required for the right section } 25 \text{ cm}$

$$\Phi_1 \left[\frac{\ell_1}{a_1 \mu_0 \mu_{r1}} \right] = \Phi_2 \left[\frac{\ell_2}{a_2 \mu_0 \mu_{r2}} \right]$$

$$\Phi_1 \ell_1 = \Phi_2 \ell_2$$

$$\Phi_1 = \Phi_2 \frac{\ell_2}{\ell_1} = 2 \times 10^{-3} \times \frac{25 \times 10^{-2}}{15 \times 10^{-2}} = 3.33 \text{ mWb}$$

$$\text{Total flux} = \Phi = \Phi_1 + \Phi_2 = 3.33 \times 10^{-3} + 2 \times 10^{-3} = 5.33 \times 10^{-3} \text{ Wb}$$

Step-3: To calculate total MMF required for left section and middle section

$$\text{MMF required for the left section } 25 \text{ cm} = \Phi \left[\frac{\ell}{a \mu_0 \mu_{r1}} \right]$$

$$= 5.33 \times 10^{-3} \left[\frac{25 \times 10^{-2}}{4 \times 10^{-4} \times (4\pi \times 10^{-7}) 600} \right] = 4420 \text{ AT}$$

$$\text{MMF required for the middle section } 15 \text{ cm} = \Phi_1 \left[\frac{\ell_1}{a_1 \mu_0 \mu_{r1}} \right]$$

$$= 3.33 \times 10^{-3} \left[\frac{15 \times 10^{-2}}{4 \times 10^{-4} \times (4\pi \times 10^{-7}) \times 600} \right] = 1657 \text{ AT}$$

Step-4: Total MMF required

Total MMF required

= MMF required for the left section 25cm+ MMF required for the middle section 15cm
or

= MMF required for the left section 25cm+ MMF required for the right section 25cm

Total MMF required

= MMF required for the left section 25cm+ MMF required for the middle section 15cm
=4420+1657

Total MMF required =6077

Step-5: To determine magnetizing current required

NI=6077

500I=6057

I=6057/500

I=12.11A

Hysteresis & B-H Curve Concepts:

Hysteresis, in the context of magnetism, describes the phenomenon where a material's magnetization lags behind the applied magnetic field. The B-H curve (Fig.5.6), or hysteresis loop, is a graphical representation of this relationship, plotting magnetic flux density (B) against magnetic field strength (H). This curve provides valuable insights into a material's magnetic properties, including its ability to retain magnetization and its resistance to demagnetization.

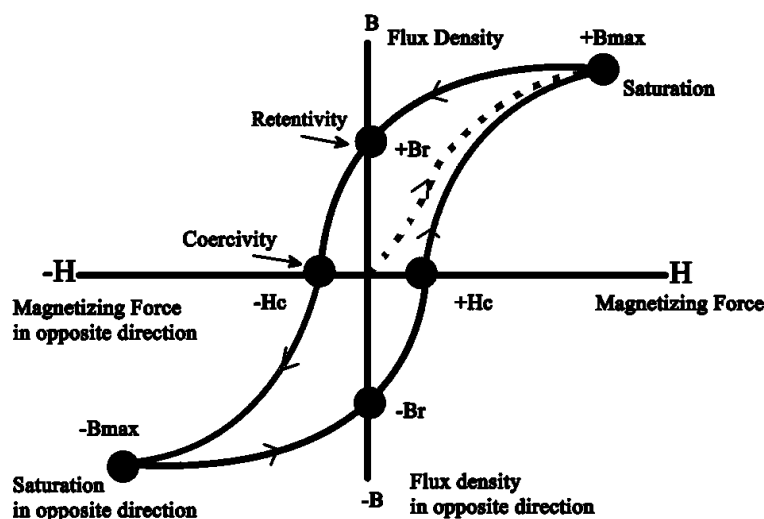


Fig.5.6.BH-Curve

When a ferromagnetic material is subjected to a changing magnetic field, its magnetization doesn't instantly respond to the field's strength. Instead, it lags behind, meaning the material retains some degree of magnetization even when the field is removed.

The B-H curve is a visual representation of this lagging effect. As the magnetic field (H) is applied and cycled, the resulting magnetic flux density (B) traces a loop. This loop is characteristic of the material and its magnetic behaviour and is known as **Hysteresis Loop**:

5.4 Faraday's laws of Electromagnetic Induction (EMI):

There are two law's of EMI:

Faraday's First Law: When the magnetic flux passing through a coil changes, an EMF is induced in the coil. This change in magnetic flux can be caused by moving a magnet towards or away from the coil, changing the strength of the magnetic field, or changing the orientation of the coil relative to the magnetic field.

Faraday's Second Law: The magnitude of the induced EMF is directly proportional to the rate of change of magnetic flux. In other words, the faster the magnetic flux changes, the greater the induced EMF.

Mathematically, $e = -\frac{d(N\Phi)}{dt}$ Volts

Negative sign is due to Lenz's law which states that the induced current will flow in a direction that opposes the change in magnetic flux that produced it.

5.5 Statically induced emf:

5.5.1. Self induced EMF:

Self-induced emf, also known as back emf, is the electromotive force (voltage) induced in a coil due to the changing magnetic flux produced by the current flowing through the coil itself. This phenomenon occurs when the current in a circuit changes, causing a corresponding change in the magnetic field, which then induces a voltage opposing the change in current as in Fig.5.7

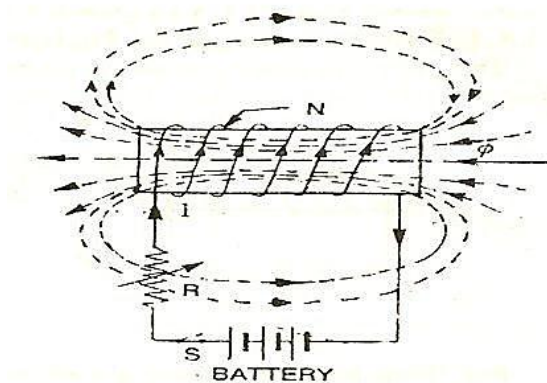


Fig.5.7: Concept of self-induced EMF

5.5.2. Mutually induced EMF:

Mutually induced EMF refers to the phenomenon where a changing current in one coil induces an electromotive force (EMF) in a nearby coil. This occurs due to the magnetic field produced by the first coil linking with the second coil, and a change in the magnetic flux induces an EMF in the second coil. The magnitude of the induced EMF is proportional to the rate of change of current in the first coil and the mutual inductance between the two coils as in Fig.5.8

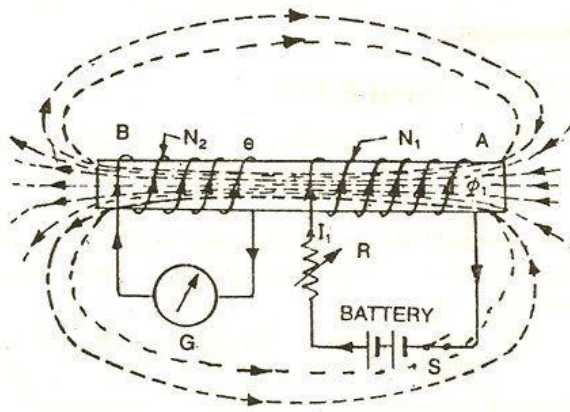


Fig.5.8: Concept of mutually-induced EMF

Example: Transformer

5.5.2. Transformer:

Definition: It is a static electrical machine which transfers power from primary winding to secondary winding without change in frequency. In a transformer, voltage can be stepped up or stepped down. The basic principle of the transformer is mutual induction

Construction of transformer:

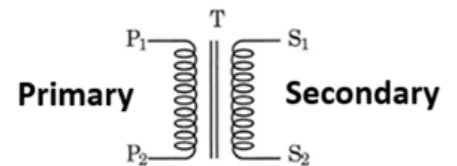
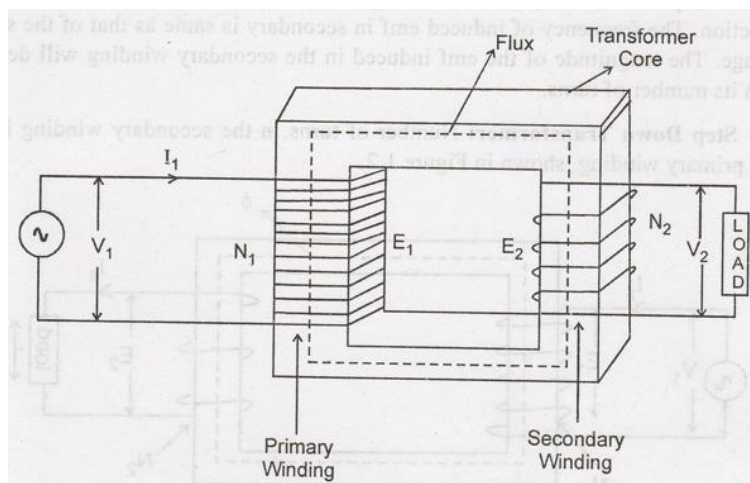


Fig.5.9: Transformer

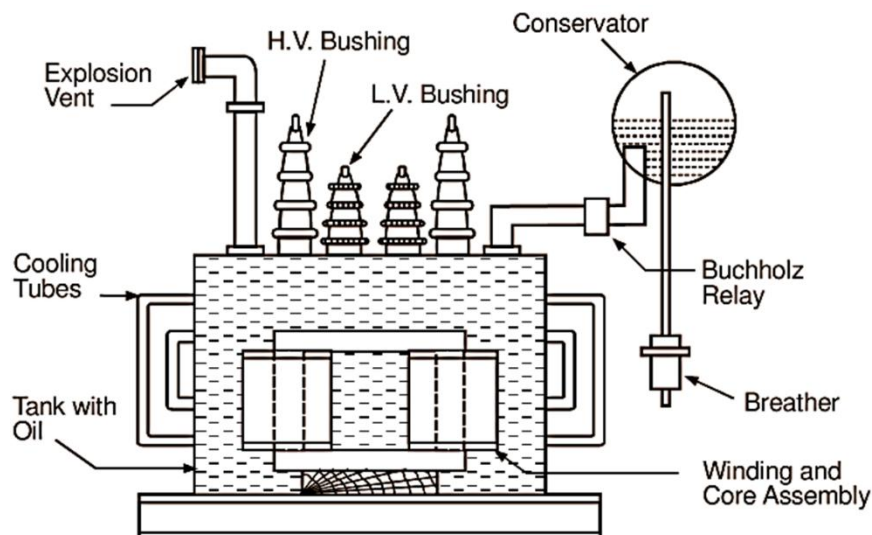


Fig.5.10: Practical Transformer

There are three major components of a Transformer:

- Iron Core

- Primary Winding
- Secondary Winding

Core: The core of the transformer is rectangular in shape and laminated. It provides a low-resistance path for the magnetic flux and minimizes energy loss due to eddy currents. Laminated sheets are stacked to minimize air gaps and reduce reluctance. They are often E and I shaped, or in a stepped construction to maximize space utilization.

Windings: There are two windings on the transformer i.e. Primary Winding and Secondary Winding. The Primary Winding is connected to the input terminal and is responsible for generating a self-induced EMF. The Secondary Winding is connected to the output load. These windings are placed on the core and are electrically insulated from each other and the core for proper functioning and reduction in losses.

These coils have different numbers of turns compared to each other. The Primary Winding of the Transformer has N_1 turns. Similarly, the Secondary Winding of the Transformer has N_2 turns. The relationship between the Primary side voltage and number of turns to that of voltage is given by $\frac{V_1}{V_2} = \frac{N_1}{N_2}$

If $N_1 > N_2$ - Step down transformer- Steps down the voltage

If $N_1 < N_2$ - Step up transformer- Steps up the voltage

Transformer is usually placed inside a tank filled with oil. Bushings are provided for the insulated connections for the windings.

Types of Transformer:

Core-type transformers have windings on the core limbs, while shell-type transformers have the core surrounding the windings. The main differences are their construction, with core types having two limbs and shell types having three, and their applications: core types are used for high voltage, while shell types are used for low voltage.

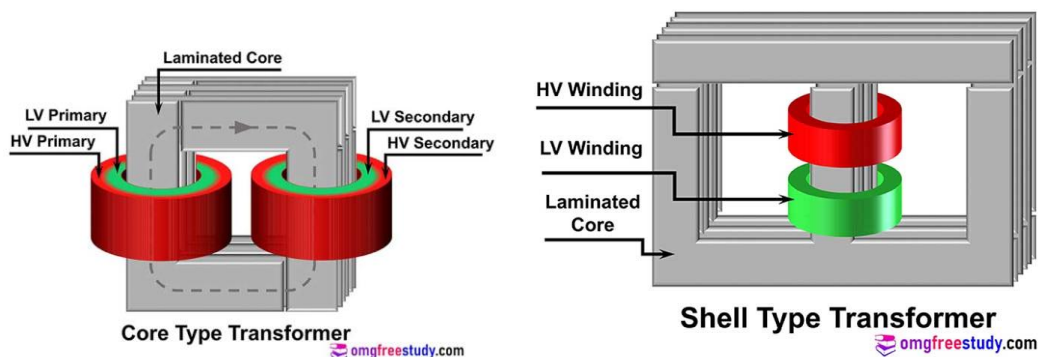


Fig.5.11 Types of Transformer

EMF equation:

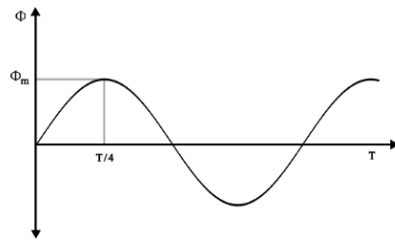
Let,

N_1 be the number of turns in the primary winding.

N_2 be the number of turns in the secondary winding.

ϕ_m be the maximum flux in core in *webers*.

f be the frequency of AC input voltage in *Hertz*.



EMF induced in the transformer winding

As shown in fig, flux ϕ_m increases from zero value to maximum value in one-quarter of the cycle, in $1/4f$ second.

$$\text{Average rate of change of flux} = \frac{\phi_m}{1/4f} = 4\phi_m f$$

By Faradays law of electromagnetic induction, rate of change of flux is nothing but induced emf,

$$\text{Average emf/turn} = 4\phi_m f$$

If the flux varies sinusoidally, then the rms value of induced emf is obtained by multiplying the average value with form factor which is 1.11.

$$\text{RMS value of emf/turn} = 1.11 \times 4\phi_m f = 4.44\phi_m f$$

The rms value of induced emf in the primary winding is given by,

$$E_1 = 4.44\phi_m f N_1 \text{ Volts}$$

Similarly, rms value of induced emf in the secondary winding is given by,

$$E_2 = 4.44\phi_m f N_2 \text{ Volts}$$

Exercise 3:

A sinusoidal flux 0.2 Wb (max.) links with 55 turns of a transformer secondary coil. Calculate the r.m.s. value to the induced EMF in the secondary coil. The supply frequency is 50 Hz.

Solution:

Given: $\phi_m = 0.2 \text{ Wb}$, $N_2 = 55$, $E_2 = ?$, $f = 50 \text{ Hz}$

$$E_2 = 4.44\phi_m f N_2 = 4.44 \times 0.2 \times 50 \times 55 = 2442 \text{ V}$$

5.5.3. Dynamically induced EMF:

Dynamically induced EMF, also known as motional EMF, is the electromotive force (EMF) generated in a conductor when it moves through a magnetic field, causing a change in magnetic flux linkage. This phenomenon is a core principle behind electrical generators and motors as in Fig.5.12

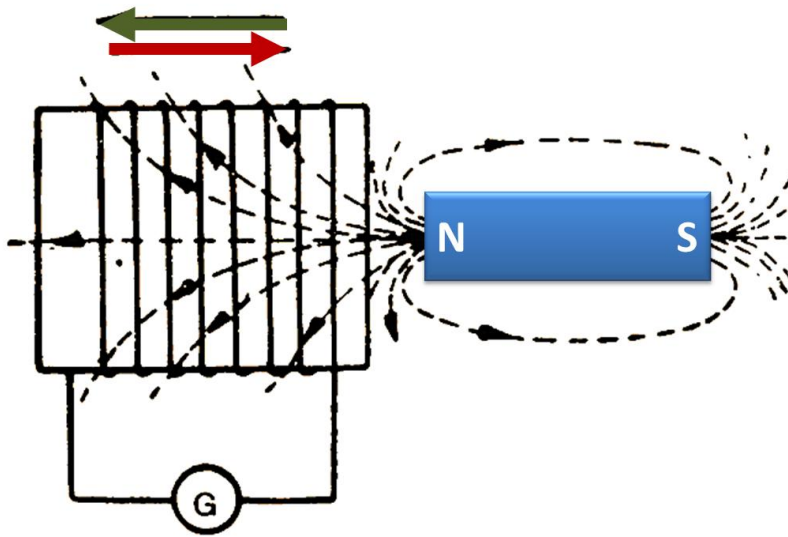


Fig.5.12: Concept of dynamically induced EMF

Example: DC Generator and DC Motor

Fleming's right-hand rule: Fleming's right-hand rule helps determine the direction of induced current in a conductor moving within a magnetic field. It's a tool primarily used in generators to understand how electricity is produced.

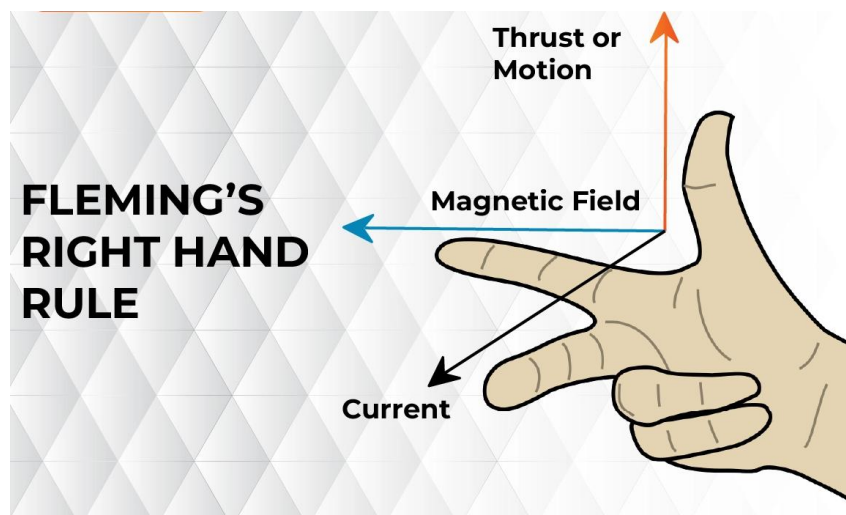


Fig.5.13. Fleming's Right Hand Rule

When three fingers i.e thumb, forefinger, and middle finger of right hand are held mutually perpendicular (at right angles) to each other, then the **thumb points** it in the direction of the conductor's motion (relative to the magnetic field), the **forefinger points** in the direction of the magnetic field, the **middle finger** indicates the direction of the induced current in the conductor.

Fleming's left-hand rule: Fleming's left-hand rule is a mnemonic used to understand the relationship between the magnetic field, current, and force in a motor. It's a tool primarily used in motors to understand how mechanical force is produced.

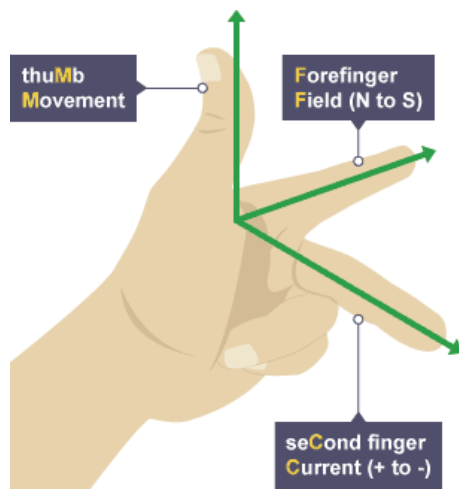


Fig.5.14. Fleming's Left Hand Rule

When three fingers i.e thumb, forefinger, and middle finger of left hand are held mutually perpendicular (at right angles) to each other, then the **index finger** points in the direction of the magnetic field, the **middle finger** points in the direction of the current, and the **thumb** will point in the direction of the force (or motion) on the conductor

5.5.4. Concept of DC Machine:

A DC machine is an **electromechanical energy conversion device**. The DC machine can be classified into two types namely DC motors and DC generators.

DC Generator: A DC generator works on the principle of electromagnetic induction. Specifically, it converts mechanical energy into direct current (DC) electrical energy by inducing a voltage in a conductor as it moves through a magnetic field. The induced voltage is dynamically induced EMF. This induced voltage then drives current through an external circuit.

DC Motor: A DC motor converts electrical energy into mechanical energy through the interaction of magnetic fields and electric current. When a current-carrying conductor is placed in a magnetic field, it experiences a force, causing rotation. This principle, governed by Lorentz force and Fleming's left-hand rule, allows a DC motor to produce torque and motion.

Construction of DC Machine:

A DC machine, either a motor or generator, has two main parts: a stator and a rotor. The stator is the stationary part, and the rotor is the rotating part. Key components include the yoke (frame), poles, field windings, armature core, armature winding, commutator, and brushes.

Stator:

1.Yoke (Frame): Provides mechanical support for other components and protects the machine from external disturbances. It also serves as a return path for the magnetic flux

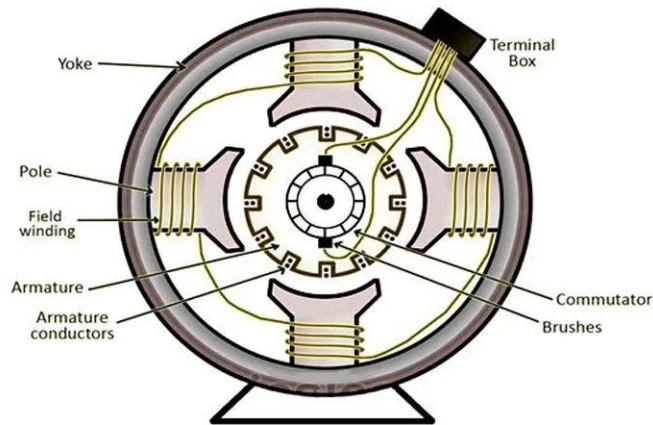


Fig.5.15: DC Machine

2.Poles and Pole Shoes: Electromagnets that produce the magnetic field. They consist of a pole core and a pole shoe. Pole shoe help to spread the flux in the air gap.

3.Field Windings: Coils of wire wound around the poles, creating a magnetic field when energized by a DC current.

4.Interpoles (Compensating Windings): Smaller poles located between main poles, used to counteract armature reaction and improve commutation.

5.Armature Core: Cylindrical structure with slots to hold the armature winding. It is made of laminated silicon steel to reduce eddy current losses.

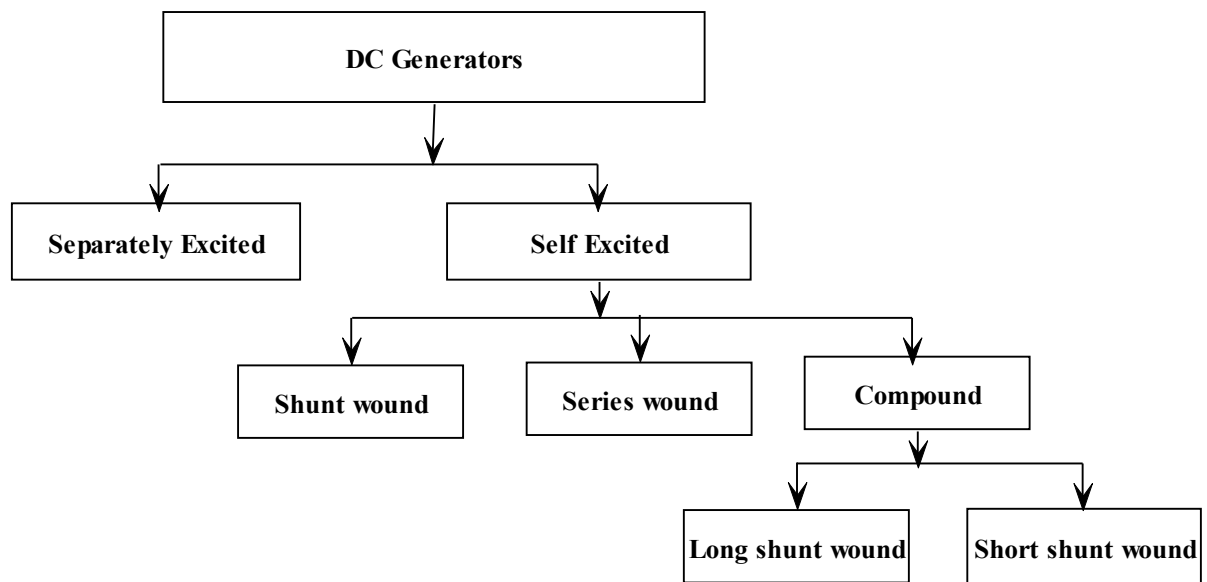
6.Armature Winding: A system of conductors where the voltage is induced.

7.Commutator: A segmented copper ring that reverses the current direction in the armature windings, ensuring unidirectional torque in motors or a steady DC output in generators.

8.Brushes: Conductive elements (usually carbon or graphite) that maintain electrical contact with the commutator to transfer current.

Types of DC Generators:

D.C. generators are generally classified according to the methods of their field excitation. The excitation of the DC machine is classified into two types namely separate excitation and self-excitation. In a separate excitation type of dc machine, the field coils are activated with a separate DC source. In the self-excitation type of dc machine, the flow of current throughout the field-winding is supplied with the machine. On the basis of these criteria, they can be classified as:



Note: There are also permanent magnet type DC generators. In these generators, no field winding is placed around the poles. These machines have fairly constant magnetic field. Although these machines are very compact but are used only in small sizes like dynamos in automobiles etc. The main disadvantages of this machine are that the flux produced by the magnets deteriorates with the passage of time which changes the characteristics of the machine.

1. Separately-excited DC generators:

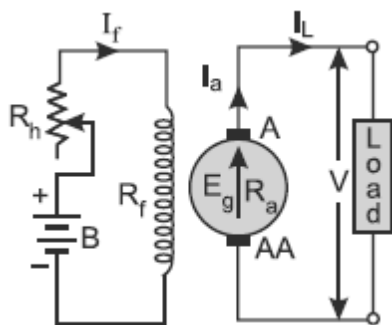


Fig.5.16. Circuit diagram for separately excited D.C. generator

A DC generator in which current is supplied to the field winding from an external DC source is called a separately excited DC generator and is shown in Fig. 5.15. The field flux remains constant.

Self-excited DC generators:

A DC generator whose field winding is excited by the current supplied by the generator itself is called a self-excited DC generator. In a self-excited DC generator, the field coils may be connected in parallel with the armature, in series with the armature or partly in series and partly in parallel with the armature winding.

Accordingly, the self-excited generators may be classified as

- (i) Shunt wound generators
- (ii) Series wound generators

(iii) Compound wound generators.

(i) Shunt wound generators:

In a shunt wound generator, the field winding is connected across the armature winding forming a parallel or shunt circuit as shown in Fig. 5.17.

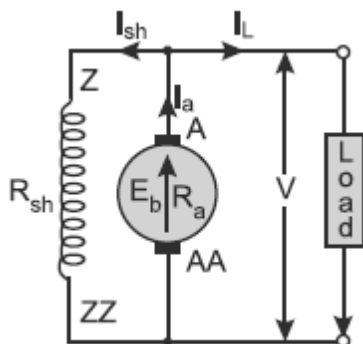


Fig.5.17. Circuit diagram for shunt D.C. generator

Therefore, full terminal voltage is applied across the field winding. A very small current I_{sh} flows through it has high resistance R_{sh}

(ii) Series wound generators:

In a series wound generator, the field winding is connected in series with the armature winding forming a series circuit and is as shown in Fig. 5.18.

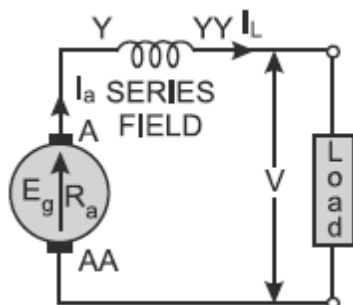


Fig.5.18. Circuit diagram for series D.C. generator

Therefore, full line current I_L or armature current I_a flows through it. Since the series field winding carries full load current, it has a few turns of thick wire having low resistance (usually of the order of less than one ohm).

(iii) Compound wound generators:

In a compound wound generator, there are two sets of field windings on each pole. One of them is connected in series (having few turns of thick wire) and the other is connected in parallel (having many turns of fine wire) with armature. A compound wound generator may be;

(a) Long shunt in which the shunt field winding is connected in parallel with the combination of both armature and series field winding. The conventional diagram of long shunt compound generator is shown in Fig. 5.19.

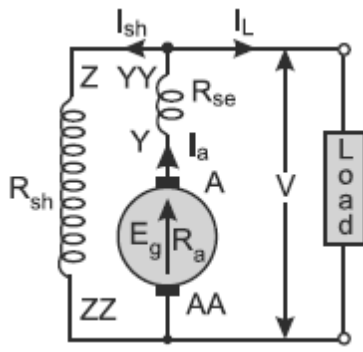


Fig.5.19. Circuit diagram for long shunt D.C. generator

Important relations

$$\text{Shunt field current, } I_{sh} = \frac{V}{R_{sh}}$$

$$\text{Series field current, } I_{se} = I_a = I_L + I_{sh} \quad (1)$$

$$\text{Terminal voltage, } V = E_g - I_a R_a - I_{se} R_{se} = E_g - I_a (R_a + R_{se})$$

$$\text{Including brush contact drop, } V = E_g - I_a (R_a + R_{se}) - 2V_b \quad (2)$$

(Note: Number of brushes is equal to number of parallel paths and if the type of winding is not given then assume number of brushes is equal to two)

$$\text{Power developed} = P_d = E_g I_a$$

$$\text{Power output} = P_{out} = V I_L$$

(b) Short shunt in which the shunt field winding is connected in parallel with only armature winding. The conventional diagram of short shunt compound generator is shown in Fig. 5.20.

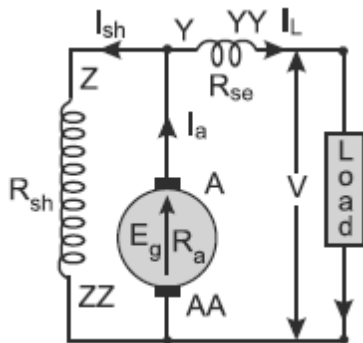


Fig.5.20. Circuit diagram for short shunt D.C. generator

Important relations

$$\text{Series field current, } I_{se} = I_L$$

$$\text{Shunt field current, } I_{sh} = \frac{V + I_{se} R_{se}}{R_{sh}}$$

$$I_a = I_L + I_{sh}$$

$$\text{Terminal voltage, } V = E_g - I_a R_a - I_L R_{se}$$

$$\text{Including brush contact drop, } V = E_g - I_a R_a - I_L R_{se} - 2V_b \quad (2)$$

(Note: Number of brushes is equal to number of parallel paths and if the type of winding is not given then assume number of brushes is equal to two)

$$\text{Power developed} = P_d = E_g I_a$$

$$\text{Power output} = P_{out} = VI_L$$

Cumulatively and differentially compound-wound generators:

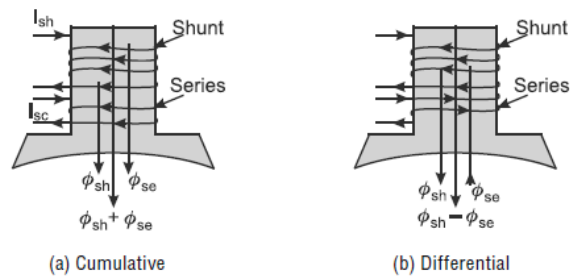


Fig.5.21. Winding position and direction of flow of current in shunt and series winding

In compound wound DC generators, the field is produced by the shunt as well as series winding. Generally the shunt field is stronger than the series field. When the series field assists the shunt field, the generator is called as **cumulatively compound wound generator** [see Fig. 5.21(a)]. However, when the series field opposes the shunt field, the generator is known as **differentially compound wound generator** [see Fig. 5.21(b)].

Types of DC motors:

On the basis of the connections of armature and their field winding, DC motors can be classified as;

1. Separately excited DC motor:

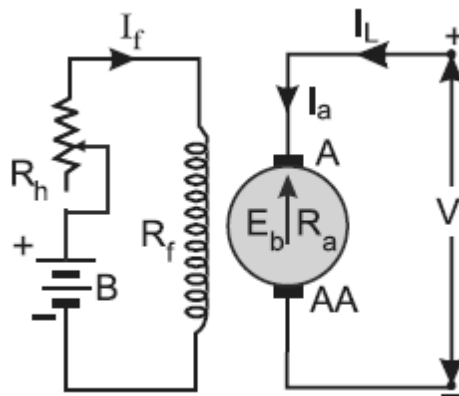


Fig.5.22. Separately excited DC motor -Circuit diagram

The conventional diagram of a separately excited DC motor is shown Fig. 5.21. Its voltage equation will be; $E_b = V - IR_a - 2V_b$ (Where) where

E_b = Back emf produced by the armature and is given by $E_b = \frac{\phi Z N}{60} \times \frac{P}{A}$ where

ϕ = armature flux in Wb, Z = number of conductors ,

A = number of parallel paths = 2 for wave and P for lap winding machine

V = Applied voltage

R_a = Armature resistance in Ω

V_b = voltage drop per brush in Volts

2. Self excited DC motors:

These motors can be further classified as;

(i) **Shunt motor:** In Shunt wound DC Motor, the field windings are in parallel with armature as shown in Fig. 5.23.

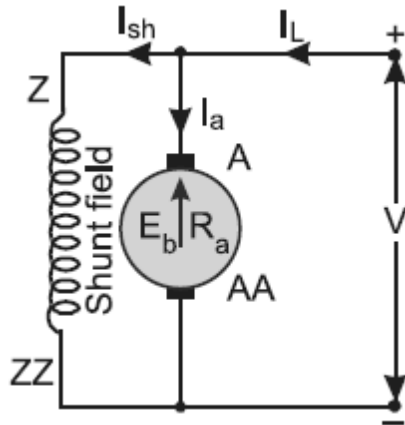


Fig.5.23 Self excited DC shunt motor -Circuit diagram

Equations governing the operation of a shunt motor are

$$I_{sh} = \frac{V}{R_{sh}}$$

$$I_a = I_L - I_{sh}$$

$$E_b = V - IR_a - 2V_b$$

Where

I_{sh} = the field current

I_L = the line current

R_{sh} = the field resistance in Ω

R_a = Armature resistance in Ω

E_b = Back emf produced by the armature and is given by $E_b = \frac{\phi Z N}{60} \times \frac{P}{A}$ where

ϕ = armature flux in Wb, Z = number of conductors ,

A = number of parallel paths = 2 for wave and P for lap winding machine

V = Applied voltage

V_b = voltage drop per brush in Volts

(ii) **Series motor:** In series-wound DC Motor, the field windings are in series with the armature and is as shown in Fig. 5.24

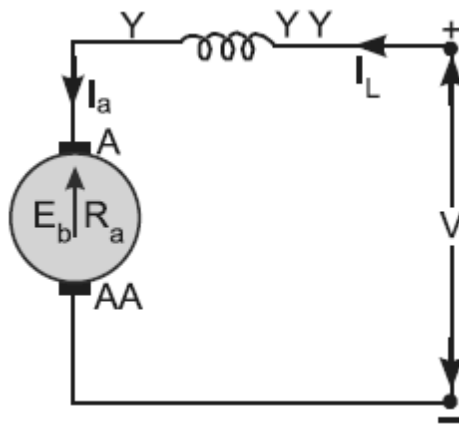


Fig.5.24. Self-excited DC series motor -Circuit diagram

Equations governing the operation of a series motor are

$$I_L = I_a = I_{se}$$

$$E_b = V - I(R_a + R_{se}) - 2V_b$$

Where

I_{se} = the field current

I_L = the line current

R_{se} = the field resistance in Ω

R_a =Armature resistance in Ω

E_b = Back emf produced by the armature and is given by $E_b = \frac{\phi Z N}{60} \times \frac{P}{A}$ where

ϕ = armature flux in Wb, Z = number of conductors ,

A = number of parallel paths = 2 for wave and P for lap winding machine

V = Applied voltage

V_b = voltage drop per brush in Volts

Exercise 4: problem on shunt motor

A 220V DC shunt motor has an armature resistance of 0.5Ω and shunt field resistance of 110Ω . If the full load line current is 20 A, find the induced back EMF, input power and the Power developed

Solution:

Step-1: Note down the given data

Given: $V=220V$, $R_a=0.5\Omega$, $R_{sh}=110\Omega$, $I_L=20A$, E_b , P_{in} , P_d

Step-2: Determine shunt field current and armature current

$$\text{Shunt field current} = I_{sh} = \frac{V}{R_{sh}} = \frac{220}{110} = 2A$$

$$\text{Armature current} = I_a = I_L - I_{sh} = 20 - 2 = 18A$$

Step-3: Determine back EMF

$$\text{Back EMF} = E_b = V - I_a R_a = 220 - 18 \times 0.5 = 211V$$

Step-4: Determine input and output powers

$$\text{Power input} = P_{in} = V I_L = 220 \times 20 = 4400W$$

$$\text{Power developed} = P_d = E_b I_a = 211 \times 18 = 3798W$$

Exercise 5: problem on series motor

A 220V DC series motor has an armature resistance of 1Ω and series field resistance of 0.5Ω . If the full load line current is 30 A, find the induced back EMF, input power and the Power developed

Solution:

Step-1: Note down the given data

Given: $V=220\text{V}$, $R_a=1\Omega$, $R_{se}=0.5\Omega$, $I_a=I_{se}=I_L=30\text{A}$, E_b , P_{in} , P_d

Step-2: Determine back EMF

$$\text{Back EMF} = E_b = V - I_a R_a - I_{se} R_{se} = 220 - 30 \times 1 - 30 \times 0.5 = 175\text{V}$$

Step-4: Determine input and output powers

$$\text{Power input} = P_{in} = VI_L = 220 \times 30 = 6600\text{W}$$

$$\text{Power developed} = P_d = E_b I_a = 175 \times 30 = 5250\text{W}$$

(iii) Compound Wound Motor

A DC Motor having both shunt and series field windings is called a **Compound Motor**. The connection diagram of the compound motor is shown below:

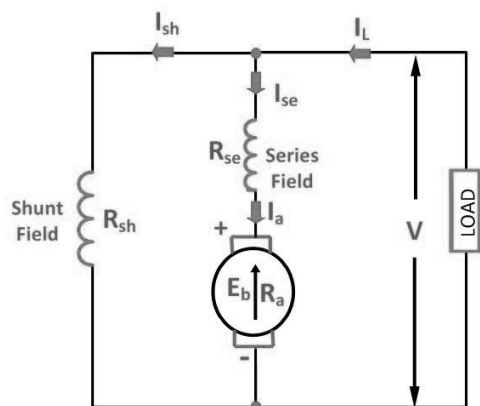


Fig.5.25. DC Compound motor-Circuit diagram

The compound motor is further subdivided as **Cumulative Compound Motor** and **Differential Compound Motor**.

In a cumulative compound motor the flux produced by both the windings is in the same direction, i.e. $\phi_r = \phi_{sh} + \phi_{se}$

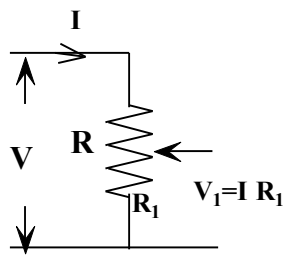
In differential compound motor, the flux produced by the series field windings is opposite to the flux produced by the shunt field winding, i.e. $\phi_r = \phi_{sh} - \phi_{se}$.

The positive and negative sign indicates that the direction of the flux produced in the field windings.

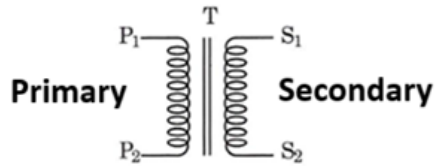
5.6.Coupled Circuits

Two circuits are said to be 'coupled' when energy transfer takes place from one circuit to the other when one of the circuits is energized. There are many types of Coupled Circuits

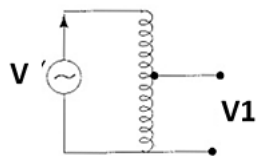
- (i) conductive coupling. Example: potential divider



- (ii) inductive or magnetic coupling Example: two winding transformers



- (iii) Both conductive and inductive coupling Example: auto transformer



5.6.1 Magnetically Coupled Circuits:

When two coils are placed very close to each other, a changing flux in one coil induces voltage in second coil. Two coils are said to be mutually coupled coils.

The coefficient of coupling (denoted by 'k') quantifies the degree of magnetic coupling between two coils. It represents the fraction of magnetic flux produced by one coil that links with the other coil. A coefficient of 1 (or 100%) indicates perfect coupling where all flux links, while 0 means no flux is linked. Consider the two coils having self-inductances L_1 and L_2 and mutual inductance M as shown in Fig 5.26.

Let

ϕ_1 be the total flux in the coil 1 in Wb

ϕ_2 be the total flux in the coil 2 in Wb

ϕ_{12}, ϕ_{21} be the linking flux between coil 1 and coil 2 in Wb

M_{12}, M_{21} are the mutual inductance between coil 1 and coil 2 = M

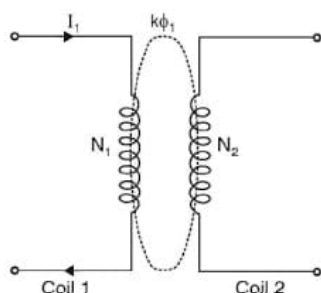


Fig.5.26: Coupling coils

Coefficient of coupling (k) is defined as the ratio of linking flux to total flux

$$\text{Coefficient of coupling} = k = \frac{\phi_{12}}{\phi_1} = \frac{\phi_{21}}{\phi_2}$$

Expression for coefficient of coupling (k):

$$\text{Coefficient of coupling} = k = \frac{\phi_{12}}{\phi_1} = \frac{\phi_{21}}{\phi_2}$$

$$\text{Mutual inductance of coil 2, } M_{12} \quad M_{12} = M = \frac{N_2 \phi_{12}}{i_1} = \frac{N_2 k \phi_1}{i_1} \quad M = \frac{N_2 \phi_{12}}{i_1} = \frac{N_2 k \phi_1}{i_1}$$

$$\text{Mutual inductance of coil 1, } M_{21} \quad M_{21} = M = \frac{N_1 \phi_{21}}{i_2} = \frac{N_1 k \phi_2}{i_2} \quad M = \frac{N_1 \phi_{21}}{i_2} = \frac{N_1 k \phi_2}{i_2}$$

$$\text{Self-inductance of coil 1, } L_1 = \frac{N_1 \phi_1}{i_1}$$

$$\text{Self-inductance of coil 2, } L_2 = \frac{N_2 \phi_2}{i_2}$$

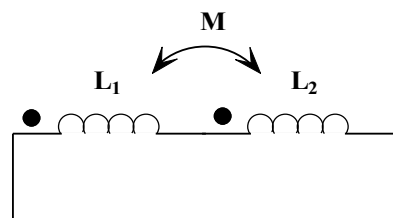
$$M_{12} * M_{21} = M * M = \frac{N_2 k \phi_1}{i_1} * \frac{N_1 k \phi_2}{i_2} = k^2 \frac{N_1 \phi_1}{i_1} * \frac{N_2 \phi_2}{i_2} = k^2 L_1 L_2 \quad M_{12} * M_{21} = M * M$$

$$= \frac{N_2 k \phi_1}{i_1} * \frac{N_1 k \phi_2}{i_2} = k^2 \frac{N_1 \phi_1}{i_1} * \frac{N_2 \phi_2}{i_2} = k^2 L_1 L_2$$

$$k = \frac{M}{\sqrt{L_1 L_2}} \quad k = \frac{M}{\sqrt{L_1 L_2}}$$

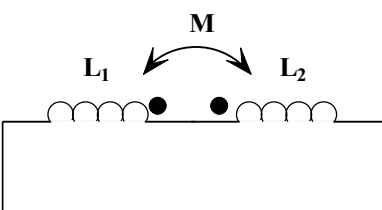
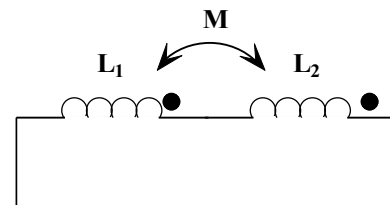
Dot convention:

The dot convention in coupled circuits indicates the polarity of induced voltage in magnetically coupled coils. It's a visual aid, using dots on the coils, to determine whether the mutual flux adds or subtracts, thus influencing the direction of induced currents. Consider the four cases shown in Fig 5.25



Total inductance of the circuit

$$L = L_1 + M + L_2 + M = L_1 + L_2 + 2M$$



Total inductance of the circuit

$$L = L_1 - M + L_2 - M = L_1 + L_2 - 2M$$

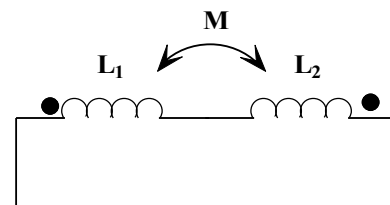


Fig.5.27 Dot conventions

Exercise 6: Find the equivalent inductance of the circuit shown below

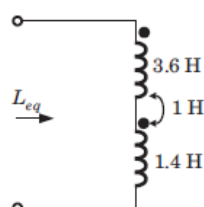


Fig.5.28 Exercise 6

$$L_{eq} = 3.6 + 1 + 1.4 + 1 = 7 \text{ H}$$

Exercise 7: Find the equivalent inductance of the circuit shown below

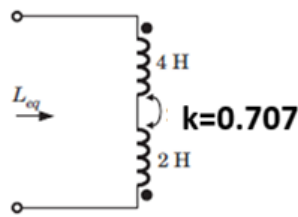


Fig.5.29 Exercise 7

$$M = k\sqrt{L_1 L_2}$$

$$M = 0.707\sqrt{4 \times 2} = 2 \text{ H}$$

$$L_{eq} = 4 - 2 + 2 - 2 + 1 = 2 \text{ H}$$

Exercise 8: Calculate the coefficient of coupling between two coils having self-inductances of 50 mH and 60 mH. The mutual inductance between them is 30 mH

$$L_1 = 50 \text{ mH}, L_2 = 60 \text{ mH}, M = 30 \text{ mH}$$

$$k = \frac{M}{\sqrt{L_1 L_2}} \quad k = \frac{M}{\sqrt{L_1 L_2}} = \frac{30 \times 10^{-3}}{\sqrt{50 \times 10^{-3} \times 60 \times 10^{-3}}} = 0.5477$$

5.6.2 Mesh analysis of coupled circuits:

Exercise 9: Two coupled coils with $L_1 = 0.5 \text{ H}$ and $L_2 = 4 \text{ H}$ have a coefficient of coupling 0.8. Find the maximum value of inductance induced EMF in the coil-2 if a current of $i_1 = 20 \sin 314t$ passed flows in the coil-1

Step1: Determine mutual inductance

$$M = k\sqrt{L_1 L_2} \quad M = k\sqrt{L_1 L_2} = 0.8\sqrt{0.5 \times 4} = 1.13$$

Step2: Determine instantaneous mutually induced EMF in coil-2

$$\begin{aligned} e_2 &= M \frac{di_1}{dt} = 1.13 \frac{d(20 \sin 314t)}{dt} = 1.13(314 \times 20 \cos 314t) = 7096.2 \cos 314t \text{ e}_2 \\ &= M \frac{di_1}{dt} = 1.13 \frac{d(20 \sin 314t)}{dt} = 1.13(314 \times 20 \cos 314t) \\ &= 7096.2 \cos 314t \end{aligned}$$

Step3: Obtain the maximum value of inductance EMF

The maximum value of inductance EMF = 7096.2 V

Exercise 10: Find the voltage across the 10 Ω resistor for the network shown in Fig 5.30

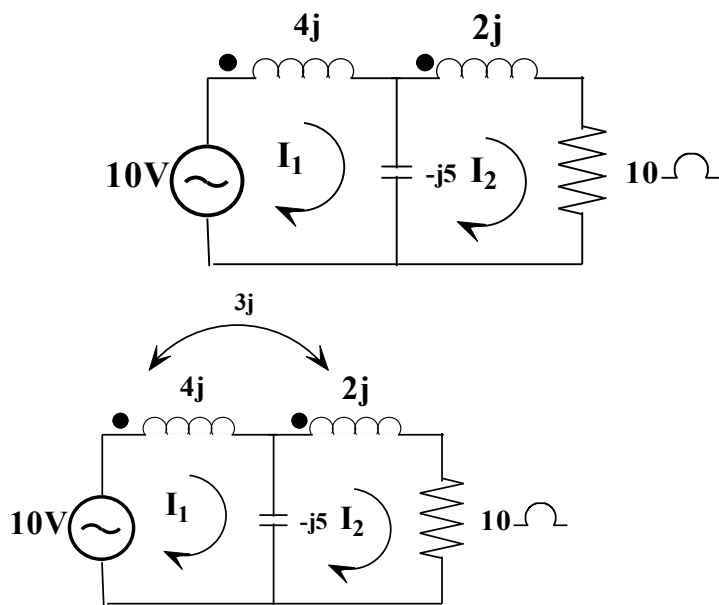


Fig 5.30 Exercise 10

Applying KVL to mesh 1,

$$-4j \times I_1 + 5j \times (I_1 - I_2) + 10 - 3j \times I_2 = 0$$

$$j \times I_1 - 8j \times I_2 = -10$$

$$-j \times I_1 + 8j \times I_2 = 10$$

Applying KVL to mesh 2,

$$-2j \times I_2 - 10 \times I_2 + 5j \times (I_2 - I_1) - 3j \times I_1 = 0$$

$$-8j \times I_1 - (10 - 3j) \times I_2 = 0$$

$$8j \times I_1 + (10 - 3j) \times I_2 = 0$$

Applying Cramm's rule,

$$I_2 = \frac{\begin{vmatrix} -j1 & 10 \\ 8j & 0 \end{vmatrix}}{\begin{vmatrix} -j1 & 8j \\ 8j & 10 - 3j \end{vmatrix}} = \frac{-80j}{61 - 10j} = 0.203 - 1.28j = 1.29 \angle -81^\circ \text{ A}$$

Voltage across 10Ω resistor is $10 \times 1.29 \angle -81^\circ = 12.9 \angle -81^\circ \text{ V}$

Exercise 11: Calculate the voltage V_2 in the circuit of Figure 5.31.

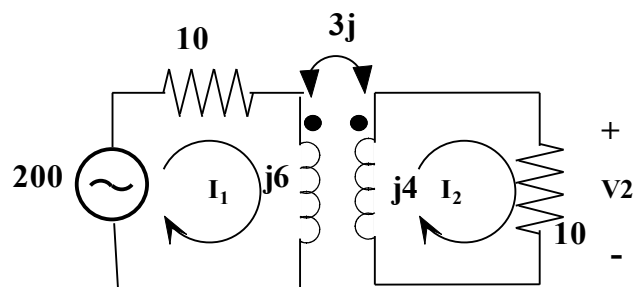


Fig 5.31 Exercise 11

Applying KVL to mesh 1,

$$-10 \times I_1 - j6 \times I_1 + 3j \times I_2 + 200 = 0$$

$$-(10 + j6) \times I_1 + 3j \times I_2 = -200$$

$$(10+j6)j\omega I_1 - 3j\omega I_2 = 200$$

Applying KVL to mesh 2,

$$-10\omega I_2 - 4j\omega I_2 + 3j\omega I_1 = 0$$

$$3j\omega I_1 - (10 + 4j)\omega I_2 = 0$$

Applying Cramm's rule,

$$I_2 = \frac{\begin{vmatrix} (10 + 6j) & 200 \\ 3j & 0 \end{vmatrix}}{\begin{vmatrix} (10 + 6j) & -3j \\ 3j & -(10 + 4j) \end{vmatrix}} = \frac{-600j}{-85 - 100j} = 3.48 + 2.96j = 4.57\angle 40.4^\circ \text{ A}$$

Voltage across 10Ω resistor is $V_2 = 10 \times 4.57\angle 40.4^\circ = 45.7\angle 40.4^\circ \text{ V}$